

Chapter 4

Dynamical systems and chaos

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4.1 Logistic map

Incredibly, an amazing complex behavior can be observed in a very simple one-dimensional difference equation:

LOGISTIC
MAP

$$x_{n+1} = f(x_n) \quad f(x) = rx(1 - x), \quad (4.1)$$

which is called the LOGISTIC MAP. Originally this map was introduced as a generalization of linear growth rx that included the damping factor $-rx^2$ caused by limited resources. In the model r is an external parameter that can be tuned both by nature or by an experimenter. The hidden complexity in (4.1) can be unveiled only with the aid of computers and May [1] was the first to realize that such a simple system is characterized by chaotic behavior. CHAOS is here defined as SENSITIVITY TO INITIAL CONDITIONS.

A difference equation like (4.1) is solved by direct iteration:

SOLVING
THE DIF-
ERENCE
EQUATION

$$\begin{aligned} x_1 &= f(x_0) \\ x_2 &= f(x_1) = f(f(x_0)) \equiv (f \circ f)(x_0) \\ x_3 &= f(x_2) = f(f(f(x_0))) \equiv (f \circ f \circ f)(x_0) \\ &\vdots \\ x_n &= \underbrace{(f \circ \dots \circ f)}_n(x_0), \end{aligned} \quad (4.2)$$

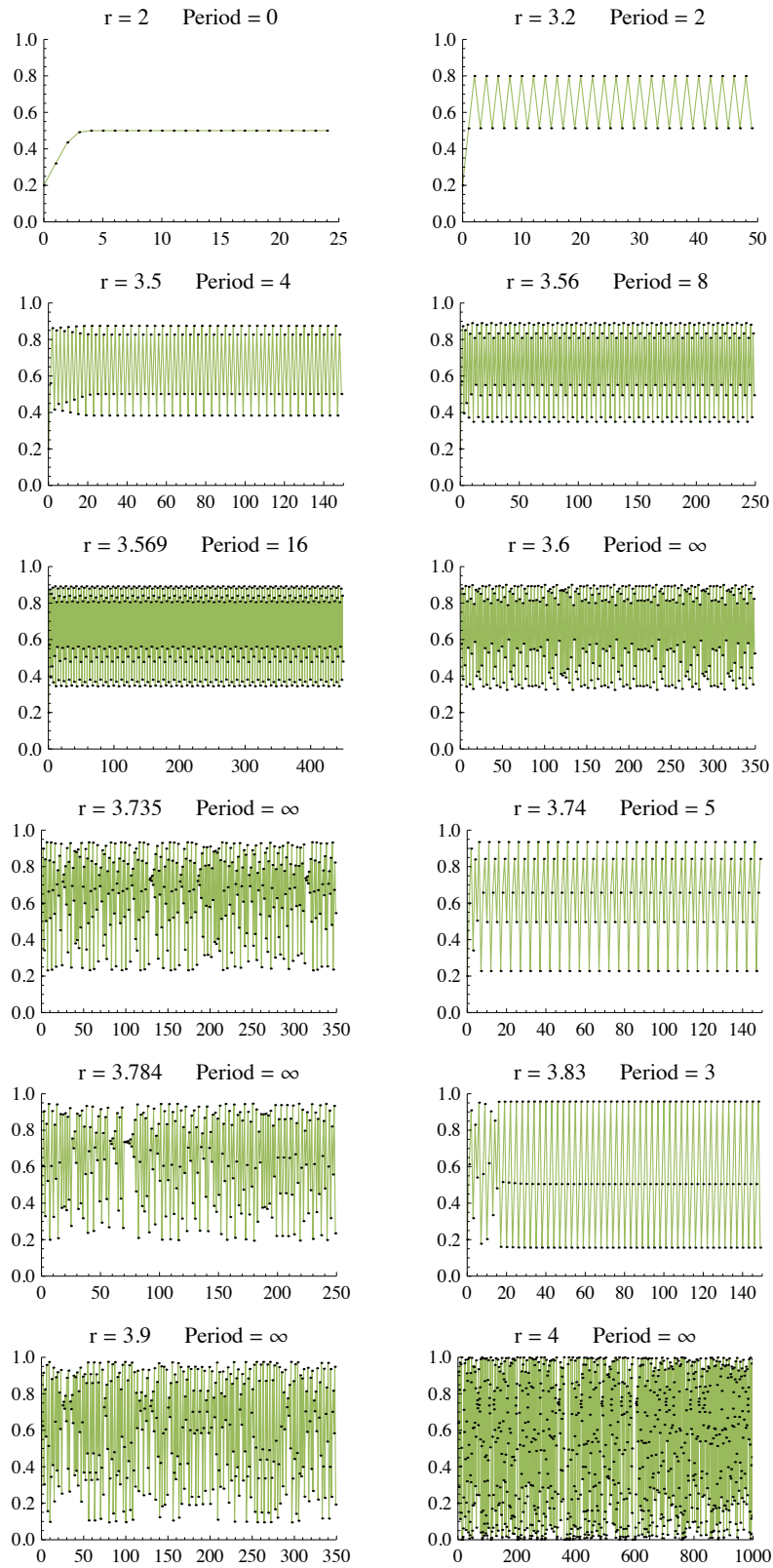


Figure 4.1: Numerical solution of the logistic map with initial condition $x_0 = 0.2$ for various values of the parameter r .

but this relation (4.2) is only formal, to understand what is going on we need to visualize the evolution. Figure 4.1 shows the solution of the logistic equation (4.1) for fixed initial condition $x_0 = 0.2$, but for various values of the parameter r .

For $r = 2$ the evolution reaches rapidly a fixed point (the same for every x_0); as we increase $r > 3$ we observe that the evolution enters a limit cycle between two fixed points (the same two for every x_0). As we increase further r the evolution goes through a sequence of period doubling bifurcations that lead to limit cycles with periods 2, 4, 8, 16, ... In particular, the values r_i at which we observe the bifurcation between the period 2^{i-1} and 2^i are:

$$\begin{aligned} r_1 &= 3 \\ r_2 &= 3.449499... \\ r_3 &= 3.544090... \\ r_4 &= 3.564407... \\ r_5 &= 3.568759... \\ &\vdots \\ r_* &= 3.5699456... \end{aligned} \tag{4.3}$$

Bifurcations accumulate at the critical value r_* , after which the behavior of the logistic map becomes aperiodic, or in other words, chaotic. The evolution of the logistic map is said to undergo a transition to chaos through a sequence of period doubling bifurcations. Looking at Figure 4.1 we see that for values of $r > r_*$ the system is chaotic (apart some windows of regularity for $r = 3.74$ and $r = 3.83$ where the period is odd): two trajectories that start near to each other diverge exponentially! This is the sensitivity to initial conditions that characterizes chaotic behavior. Finally for $r = 4$ the logistic map becomes completely random: given any initial condition, the value x_∞ is a random variable (with well defined calculable pdf) on the interval $[0, 1]$.

A better way to understand (or to see) the behavior of the logistic map is to follow the dynamics using a phase space diagram in the (x, f) plane. This is shown in Figure 4.2. Here we plot the attractor, which is the

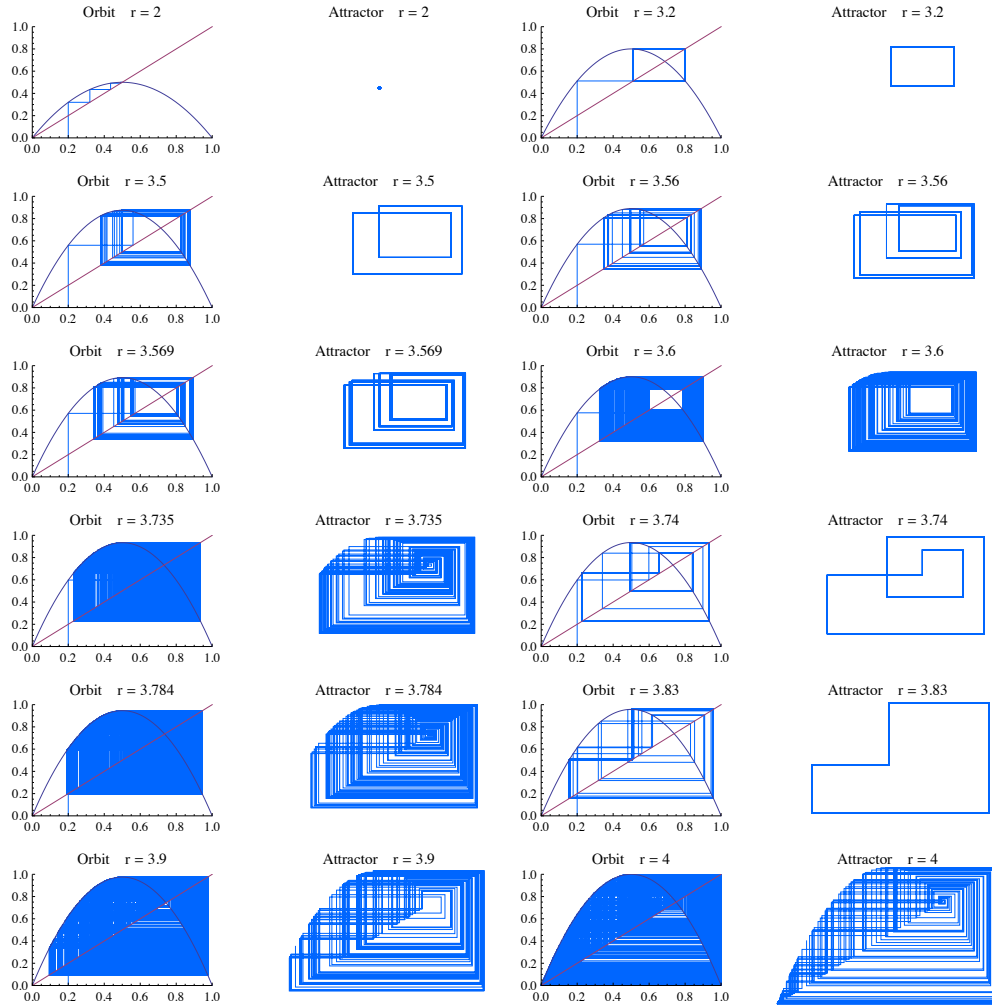


Figure 4.2: Numerical solution of the logistic map in phase space with initial condition $x_0 = 0.2$ for various values of the parameter r . For every value of r we show both the evolution and the attractor.

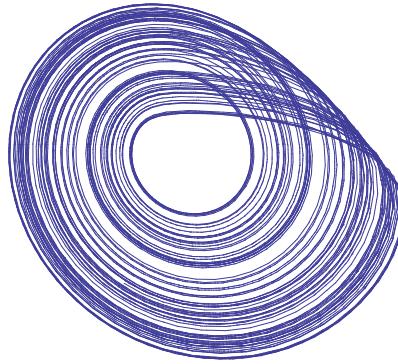


Figure 4.3: The Rossler attractor.

limit set of the evolution. When the parameter r is increased the attractor experiences bifurcations at the values r_i , and at r_* it becomes a STRANGE ATTRACTOR, i.e. the phase space locus of aperiodic trajectories. For $r = 3.74$ and $r = 3.83$ one can observe the windows of odd period regularity.

Strange attractors are also observed in continuous maps. The simplest strange attractor is obtained by solving the (dissipative) system of ordinary differential equations¹ proposed by Rossler: STRANGE ATTRACTORS

$$\begin{aligned}\dot{x} &= -y - z \\ \dot{y} &= x + ay \\ \dot{z} &= a - z(r - x),\end{aligned}$$

where a and r are parameters. Here r plays the role of the non-linearity parameter, by tuning it the system undergoes a sequence of period doubling bifurcations after which it becomes chaotic. The Rossler strange attractor is shown in Figure 4.3; one can see the similarity with the Logistic map strange attractor of Figure 4.2. Strange attractors are fractal objects and as such are interesting objects to study, but for the moment we will not pursue this.

We can construct a BIFURCATION DIAGRAM by plotting the limit points x_∞ of the dynamics generated by the logistic map as a function of the control parameter r . In this way we obtain the intriguing Figure 4.4 which BIFURCATION DIAGRAM

¹Note that a continuous system can be chaotic only if it has three variables or more.

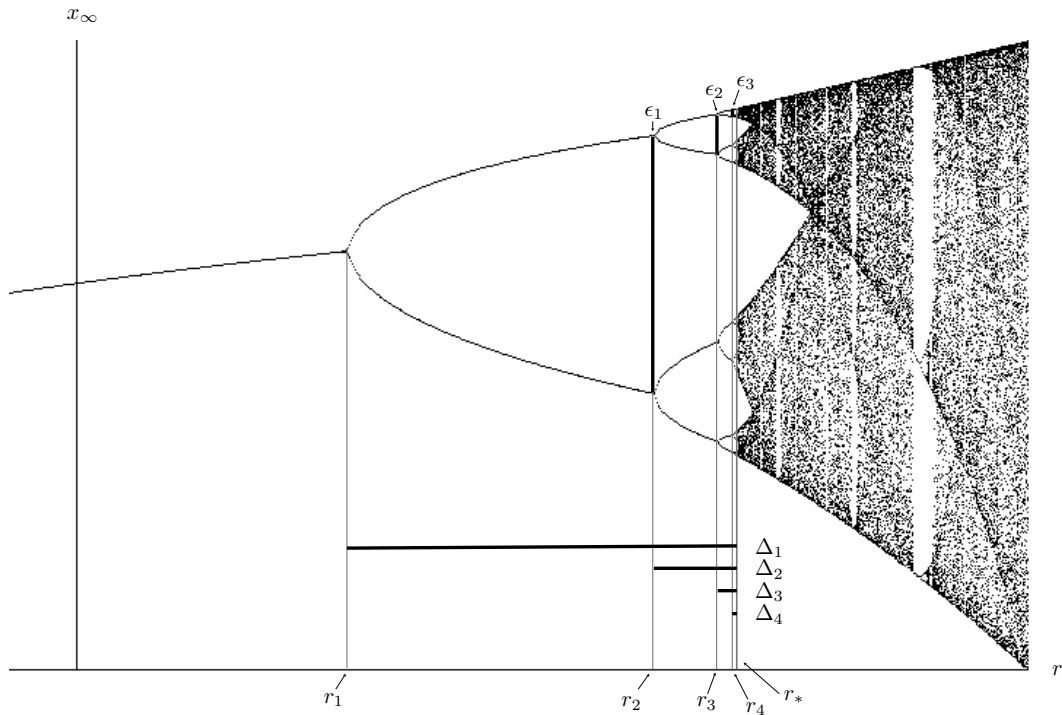


Figure 4.4: Bifurcation diagram for the logistic map.

compactly characterizes the complexity encoded in the logistic map. The diagram clearly shows that bifurcations accumulate at r_* , the chaotic regions and the odd period windows.

4.2 RG analysis of the transition to chaos

We want to see if we can understand the period doubling route to chaos using the RG analysis. In particular we want to see if universality is at work also in this case, and thus if we can extract a universal law to confront with experiments.

UNIVERSALITY
IN A
DETER-
MINISTIC
SYSTEM?

In the deterministic setting the interesting dimension is time, and thus the important scales are the time scales. To construct the RG transformation we first need to do a temporal coarse-graining. This is shown in Figure 4.5. Intuitively to temporal coarse-grain a system is like looking at it under a strobo-light tuned to show only, say, the even times positions x_{2n} . This

TEMPORAL
RG
TRANS-
FORMA-
TION

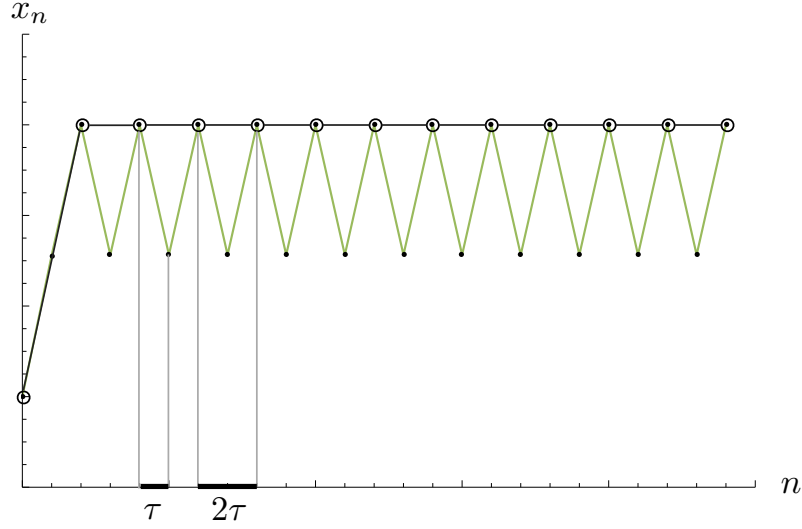


Figure 4.5: Temporal coarse-graining of the evolution of the logistic map. The coarse-graining maps a limit cycle of period 2^i to one with period 2^{i-1} .

means that our coarse-grained function is just given by composition:

$$\mathcal{C}f(x) = (f \circ f)(x). \quad (4.4)$$

More specifically, for the logistic map we have $\mathcal{C}f(x) = f(f(x)) = r^2x(1-x)(1-rx(1-x))$. The coarse-grained function is not anymore a UNIMODULAR FUNCTION (i.e. a maps with one parabolic maximum) and we need to choose the similarity transformation $\mathcal{S}$ in such a way to turn it back. We need to set:

$$\mathcal{S} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + T, \quad (4.5)$$

where the translation has to be chosen as $T = (x_*/\alpha, 0)$ to map the point (x_*, x_*) to the origin (see Figure 4.6). To simplify the similarity transformation we can make the change of variable $x \rightarrow x + \frac{1}{2}$ in (4.1) to work with maps that have their maximum at $x = 0$ (the logistic map becomes the map $f(x) = r/4 - rx^2$). If we now consider y a function of x and we define $\alpha = -1/\lambda < -1$ we have that under the similarity (4.5):

$$y(x) \xrightarrow{\mathcal{S}} \alpha y(x/\alpha). \quad (4.6)$$

Now $y = \mathcal{C}f = f \circ f$ and so the exact functional RG transformation is:

$$\mathcal{R}f(x) = \alpha f(f(x/\alpha)), \quad (4.7)$$

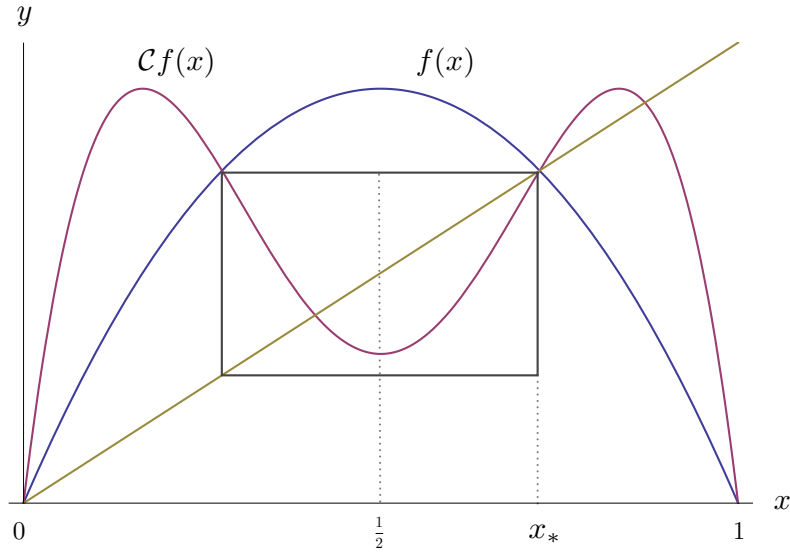


Figure 4.6: Coarse-grained function $\mathcal{C}f$ compared to the function f . The part of $\mathcal{C}f$ in the rectangle can be transformed by a similarity to a unimodal function.

with α to be determined self-consistently. This exact RG transformation was first obtained by P. Civitanovic [2].

The RG reasoning is: at larger time scales the order of the period is reduced. Since chaos is equivalent to infinite order of periodicity, the transition to chaos must be described by a RG fixed point at which the system is time-scale invariant. Thus **TRANSITION TO CHAOS** $\equiv$ **FIXED POINT**! From the RG transformation (4.7) we see that a fixed point is solution of:

$$f_*(x) = \alpha f_*(f_*(x/\alpha)). \quad (4.8)$$

If we normalize the solution as $f_*(0) = 1$, we find from (4.8) that $f_*(1) = \frac{1}{\alpha}$. The fixed point equation is a non-linear “composite” (or nested) functional equation for f_* . There is no known way to solve it exactly. We thus try to find an approximate solution by a truncation procedure [3]:

$$f_*(x) = 1 + \sum_{n=1}^{N_{tr}} c_n x^{2n}, \quad (4.9)$$

since the fixed point function is symmetric. To lowest truncation order $N_{tr} = 1$ equation (4.8) becomes:

$$1 + c_1 x^2 + O(x^4) = \alpha(1 + c_1) + \frac{2c_1^2}{\alpha} x^2 + O(x^4),$$

which can be solved by imposing:

$$\alpha(1 + c_1) = 1 \quad c_1 = \frac{2c_1^2}{\alpha},$$

from which we pick the solution which has $\alpha < -1$: $\alpha = -2.732$ and $c_1 = -1.366$. In the $N_{tr} = 2$ case we find:

$$\begin{aligned} 1 + c_1 x^2 + c_2 x^4 + O(x^6), &= \alpha(1 + c_1 + c_2) + \frac{1}{\alpha} (2c_1^2 + 4c_2 c_1) x^2 \\ &+ \frac{1}{\alpha^3} (c_1^3 + 6c_2 c_1^2 + 2c_2 c_1 + 4c_2^2) x^4 + O(x^6), \end{aligned}$$

which gives us the equations:

$$\alpha(1 + c_1 + c_2) = 1 \quad \alpha c_1 = 2c_1^2 + 4c_2 c_1 \quad \alpha^3 c_2 = c_1^3 + 6c_2 c_1^2 + 2c_2 c_1 + 4c_2^2.$$

The solutions are $c_1 = -1.522$, $c_2 = 0.128$ and $\alpha = -2.534$. When we extend the truncation to higher N_{tr} we observe convergence to the limiting value $\alpha = -2.5029...$, as shown in Figure 4.7.

By expanding the RG equation (4.7) around the fixed point,

LINEAR RG
TRANS-
FORMA-
TION

$$\begin{aligned} \mathcal{R}(f_* + \epsilon h)(x) &= \alpha f_* \left(f_* \left(\frac{x}{\alpha} \right) \right) \\ &+ \epsilon \left[\alpha h \left(\frac{x}{\alpha} \right) f'_* \left(f_* \left(\frac{x}{\alpha} \right) \right) + \alpha h \left(f_* \left(\frac{x}{\alpha} \right) \right) \right] + O(\epsilon^2) \\ &= f_*(x) + \epsilon \mathcal{L}_{f_*} h(x) + O(\epsilon^2), \end{aligned}$$

we find the linear RG operator:

$$\mathcal{L}_{f_*} h(x) = \alpha h \left(\frac{x}{\alpha} \right) f'_* \left(f_* \left(\frac{x}{\alpha} \right) \right) + \alpha h \left(f_* \left(\frac{x}{\alpha} \right) \right), \quad (4.10)$$

which acts on the functional space of unimodular functions.

We can consider now the RG eigenvalue problem [3]:

THE
EIGEN-
VALUE
PROBLEM

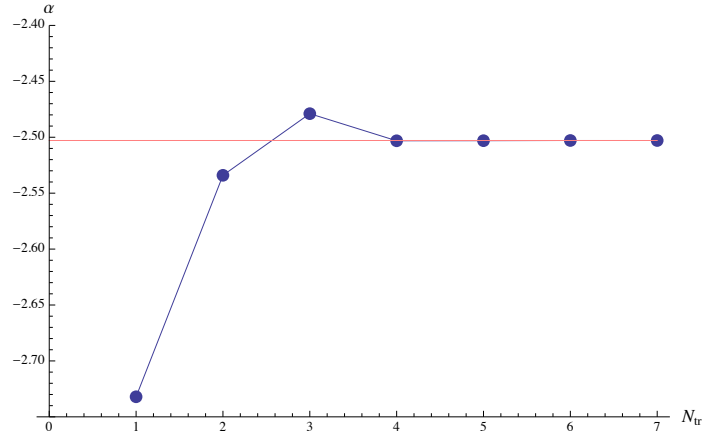


Figure 4.7: The scaling factor $\alpha(N_{tr})$ as a function of the truncation. The limiting value is $\alpha = -2.5029\dots$

$$\mathcal{L}_{f_*} h(x) = \lambda h(x). \quad (4.11)$$

As the fixed point equation, we cannot solve (4.11) exactly. We can expand the perturbation in a truncation:

$$h(x) = \sum_{n=0}^{N_{tr}} h_n x^{2n}$$

to obtain:

$$\sum_{n=0}^{N_{tr}} \left\{ \alpha \left[f_* \left(\frac{x}{\alpha} \right) \right]^{2n} + \alpha \left(\frac{x}{\alpha} \right)^{2n} f'_* \left(f_* \left(\frac{x}{\alpha} \right) \right) - \lambda x^{2n} \right\} h_n = 0.$$

A discretized version of (4.11) can be obtained by discretizing $x = \{x_0 = 0, x_1, \dots, x_m, \dots, x_{N_{tr}} = 1\}$ and defining:

$$X_{mn} = x_m^{2n} \quad L_{mn} = \alpha \left[f_* \left(\frac{x_m}{\alpha} \right) \right]^{2n} + \alpha \left(\frac{x_m}{\alpha} \right)^{2n} f'_* \left(f_* \left(\frac{x_m}{\alpha} \right) \right),$$

so that (4.11) becomes the matrix equation:

$$(\mathbb{L} - \lambda \mathbb{X}) = 0.$$

The spectrum is thus obtained by solving:

$$\det(\mathbb{L} - \lambda \mathbb{X}) = 0.$$

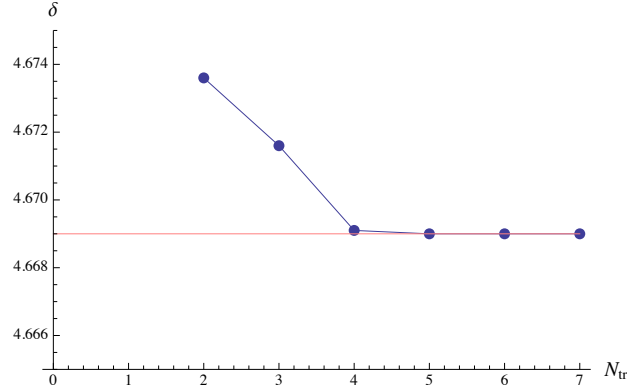


Figure 4.8: The critical exponent $\delta(N_{tr})$ as a function of the truncation. The limiting value is Feigenbaum's constant $\delta = 4.669\dots$

For $N_{tr} = 2$ we have $x = \{0, 1\}$ and:

$$\mathbb{L} = \begin{pmatrix} 3.76163 & -2.5029 \\ 2.75098 & -0.602904 \end{pmatrix} \quad \mathbb{X} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The eigenvalue equation is $\lambda^2 - 5.662\lambda + 4.617 = 0$ with solutions $\delta = \lambda_1 = 4.674$ and $\lambda_2 = 0.988$. Figure 4.8 shows that the only eigenvalue greater than one, which is Feigenbaum's constant, converges rapidly² to:

$$\delta = 4.669\dots \quad (4.12)$$

The spectrum we find has the following form:

$$\lambda_1 = \delta > \lambda_2 = 1 > \lambda_3 > \lambda_4 > \dots \quad (4.13)$$

with only one relevant and one marginal eigenperturbation.

With the understanding of the eigenvalue spectrum we are ready to understand the period doubling transition to chaos. Let f_{r_i} be a function corresponding to a dynamics with period 2^i and control parameter r_i . The RG transformations acts, by construction, by halving the period: $\mathcal{R}f_{r_i} = f'_{r'_{i-1}}$ with $f'_{r'_{i-1}}$ representing a dynamics, different from the dynamics f_{r_i} , with period 2^{i-1} and with control parameter r'_i . Theory space is foliated by the

²In fact geometrically.

infinite-dimensional bifurcation manifolds $\mathcal{M}_i$, defined to contain all functions representing dynamics at their respective i -th bifurcation. For example both f_{r_i} and $f'_{r'_i}$ belong to $\mathcal{M}_i$. The control parameters $r, r', \dots$ parametrize the one-dimensional manifolds perpendicular to the $\mathcal{M}_i$. The manifold $\mathcal{M}_\infty$ is the critical surface related to the fixed point f_* and it contains the critical dynamics $f_{r_*}, f'_{r'_*}, \dots$. We call $\mathcal{U}$ the one-dimensional unstable manifold of the fixed point f_* . See Figure 4.9 for a picture of theory space. With respect to the manifolds $\mathcal{M}_i$, we can write the RG transformations as the sum of a parallel and of an orthogonal transformation:

$$\mathcal{R} = \mathcal{R}_\perp + \mathcal{R}_\parallel,$$

defined so that:

$$\mathcal{R}_\perp f_{r_i} = f_{r_{i-1}} \in \mathcal{M}_{i-1} \quad \mathcal{R}_\parallel f_{r_i} = f'_{r'_i} \in \mathcal{M}_i. \quad (4.14)$$

The explicit construction of $\mathcal{R}_\perp$ can be found in [2]. The limit $\lim_{n \rightarrow \infty} \mathcal{R}_\parallel^n f_{r_i} = f_{*i}$ is a universal i -th bifurcation function $f_{*i} \in \mathcal{U}$. One has $\mathcal{R}_\perp f_{*i} = f_{*i-1}$ and, obviously, $\lim_{n \rightarrow \infty} \mathcal{R}_\parallel^n f_{r_*} = f_*$.

If we start from f_{r_i} , with i large so that we are near f_{r_*} , we rapidly flow towards the unstable manifold $\mathcal{U}$ and we find ourselves in a regime where the RG flow is mainly perpendicular to the manifolds $\mathcal{M}_i$. Thus we can make the approximation $\mathcal{R} \approx \mathcal{R}_\perp$, which using (4.14) means that we can write:

$$\mathcal{R} f_{r_i} \approx f_{r_{i-1}}. \quad (4.15)$$

Linearizing (4.15) near the fixed point and keeping only the relevant deformation gives:

$$\begin{aligned} f_{r_{i-1}} &\approx \mathcal{R} f_{r_i} \\ &= \mathcal{R} (f_* + (f_{r_i} - f_*)) \\ &\approx \mathcal{R} f_* + \mathcal{L}_{f_*} (f_{r_i} - f_*) + \dots \\ &\approx f_* + \delta (f_{r_i} - f_*) + \dots; \end{aligned}$$

thus:

$$f_{r_{i-1}} - f_* \sim \delta (f_{r_i} - f_*) \quad \Rightarrow \quad \delta \sim \frac{f_{r_{i-1}} - f_*}{f_{r_i} - f_*}. \quad (4.16)$$

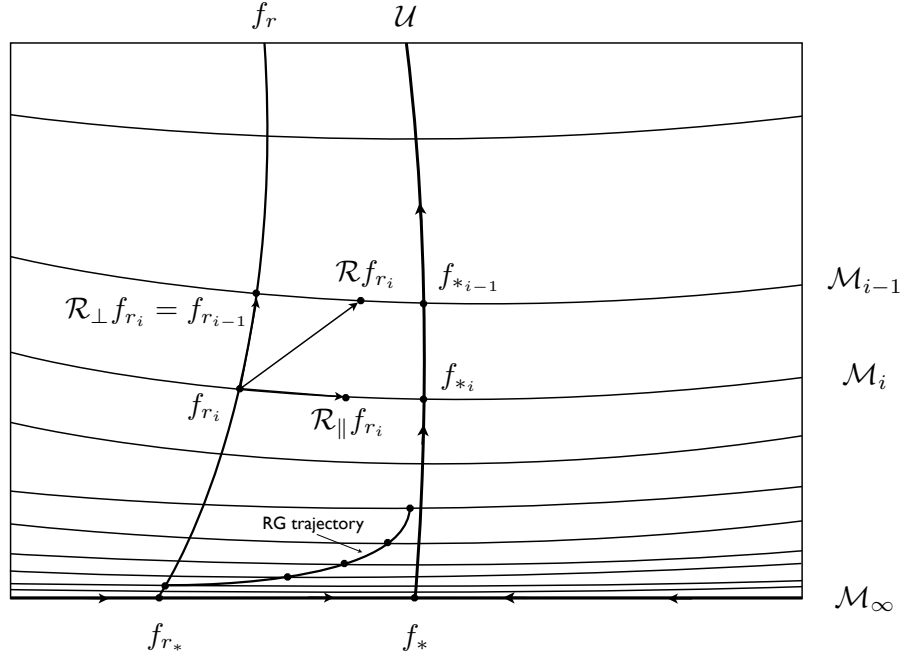


Figure 4.9: Theory space.

If we make the identification $f_{r_*} \approx f_*$, expand as $f_{r_i} = f_{r_* + (r_i - r_*)} \approx f_{r_*} + (r_i - r_*)\partial_r f_{r_*} + \dots$ and insert into (4.16), we obtain:

$$r_{i-1} - r_* \sim \delta(r_i - r_*), \quad (4.17)$$

or, with reference to the bifurcation diagram of Figure 4.4:

$$\delta \sim \frac{r_{i-1} - r_*}{r_i - r_*} = \frac{\Delta_{i-1}}{\Delta_i} \quad i \rightarrow \infty. \quad (4.18)$$

This relation shows that Feigenbaum's constant δ is related to the ratio of the distance Δ_i of successive bifurcations from r_* . Since we derived (4.18) from an RG argument we know that it represents a UNIVERSAL LAW characterizing the period doubling transition to chaos: every dynamics that belongs to our theory space will obey (4.18), i.e. the values r_i and r_* will depend on the map under consideration, but their ratio δ will not. Thus the RG eigenvalue δ is a prediction that can be tested in an experiment! WITH THE HELP OF THE RG ANALYSIS WE WERE ABLE TO FIND ORDER INTO CHAOS. Finally, we can rewrite (4.17) as:

$$r_i - r_* \sim \delta^n(r_{i+n} - r_*);$$

by imposing $j = 0$ with $r_0 = 0$ gives:

$$\frac{r_n}{r_*} \sim 1 - \frac{1}{\delta^n} \quad n \rightarrow \infty, \quad (4.19)$$

which is another form for the universal law; it describes the geometric³ convergence of the sequence of period doubling bifurcations.

We are now in the position of checking universality, i.e. we can compare our theoretical prediction (4.18) or (4.19) with the values for the r_i obtained by our numerical solution of the logistic map and shown in Figure 4.4. Using the values (4.3) we find:

$$\begin{aligned} \delta_1 &= \frac{\Delta_1}{\Delta_2} = 4.732... & \delta_2 &= \frac{\Delta_2}{\Delta_3} = 4.658... \\ \delta_3 &= \frac{\Delta_3}{\Delta_4} = 4.668... & \delta_4 &= \frac{\Delta_4}{\Delta_5} = 4.668... \end{aligned}$$

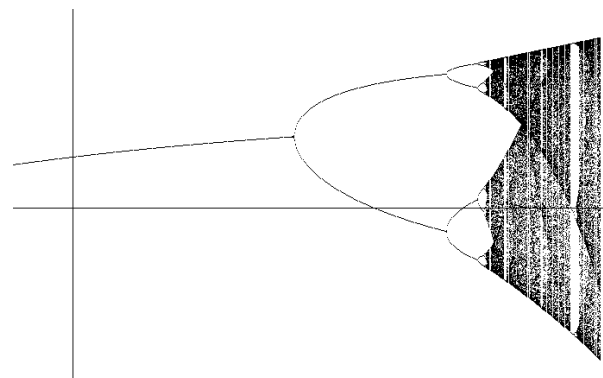
which is rapidly converging to the RG eigenvalue (4.12)! To test universality further, we can consider other maps which belong to our theory space, such as those shown in Figure 4.10. All maps exhibit the period doubling bifurcation transition to chaos. The relative location of the bifurcations r_i depends on the map, as the critical value r_* , but you can check the amazing fact that the $\delta_n = \Delta_n/\Delta_{n+1}$ tend to Feigenbaum's constant as $n \rightarrow \infty$, as indeed predicted by the RG analysis. Other more important checks can be made directly with experiments [2].

Finally, one can check that the bifurcation diagram is self-similar (affine) by enlarging it in an obvious way. In particular one can check that the ratios

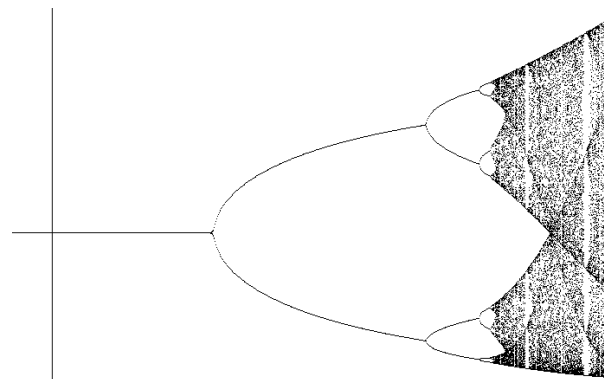
$$\alpha_i = \frac{\epsilon_{i-1}}{\epsilon_i},$$

defined in Figure 4.4, tend to the scale factor $\alpha = -2.5029...$ as $i \rightarrow \infty$; and this is true for any unimodular map!

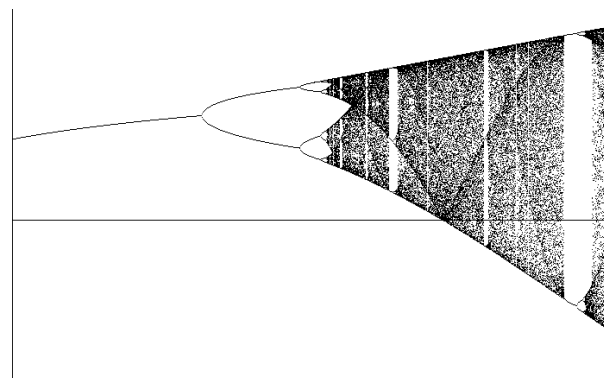
³And thus scale invariant...



$$f(x) = r/4 - rx^2$$



$$f(x) = x e^{r(1-x)}$$



$$f(x) = r \sin(\pi x)$$

Figure 4.10: Bifurcation diagrams and the relative maps.

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